How Psychology Knows Things

What the field studies, where it came from, the methods it uses to separate a real finding from a good story, and what happened when those findings were re-tested.

Isagi · November 2026 · Spine: Ciccarelli & White, *Psychology* (4th ed.), Ch. 1 — rewritten, updated and applied

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Every manager is an amateur psychologist. You decide why a dealer stopped ordering, whether a scheme will move a sluggish SKU, which salesperson is about to quit. Most of those calls run on instinct and stories. Psychology is the attempt to do the same job with evidence. This first part is about the toolkit, not the findings. Every later part — memory, persuasion, motivation, personality — leans on the methods here. If you know how a claim was tested, you know how much weight it can bear.

1 What psychology actually is

Psychology is the scientific study of **behaviour** (what can be observed: a purchase, a resignation, a reaction time) and **mental processes** (what cannot be seen directly: memory, attitudes, emotions, decisions). The word that matters is *scientific*. Folk wisdom says both "birds of a feather flock together" and "opposites attract"; it can explain any outcome after the fact, which means it predicts nothing. Psychology commits in advance and checks.

The field has four goals, and they map neatly onto management:

Goal	Question	Field-sales version
Describe	What is happening?	Order frequency fell 20% at 30 dealers in Q2.
Explain	Why is it happening?	A competitor offered longer credit; dealers are cash-constrained.
Predict	When will it happen again?	Dealers with high outstanding balances are most at risk.
Control	How can we change it?	Targeted credit terms for the at-risk group.

"Control" sounds sinister, but in psychology it just means changing behaviour deliberately — therapy, training, nudges, incentive design. Most of what a manager does falls under it.

2 Where it came from: five early schools

Psychology split off from philosophy in **1879**, when Wilhelm Wundt opened the first psychology laboratory in Leipzig. Within about 60 years it went through several competing schools, each a reaction to the last. You need them because their ideas never fully died; they survive inside modern theories and inside management thinking.

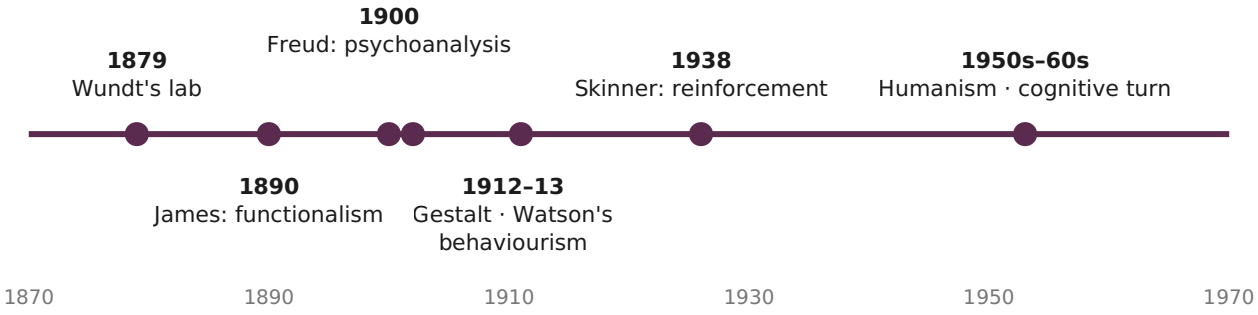


Figure 1. The founding century in one line. Dates are approximate starting points, not end points; most of these schools overlapped.

- **Structuralism** (Wundt, then Titchener): break conscious experience into its smallest parts by trained self-observation ("introspection"). It died out because introspection cannot be verified — two observers disagree and no one can settle it. Lesson: a method nobody else can check is not a method.

- **Functionalism** (William James): ask not what the mind is made of but what it is *for* — how it helps us adapt. Its descendants include industrial/organisational (I/O) psychology, the branch that studies hiring, training and work.
- **Gestalt** (Wertheimer): the whole is different from the sum of its parts. We perceive a melody, not individual notes; a brand, not a list of attributes. It returns in Part 2.
- **Psychoanalysis** (Freud): much behaviour is driven by an unconscious mind shaped by early experience. Most specific Freudian claims have not survived testing, but the broad idea that people do not fully know why they act did — it is the foundation of modern consumer research.
- **Behaviourism** (Watson, later Skinner): forget the mind; study only observable stimuli and responses. Skinner showed that behaviour followed by reinforcement is repeated. Every incentive scheme, loyalty programme and sales contest is applied behaviourism. Part 4 covers it in depth.

3 Seven modern perspectives

Today no single school rules. Psychologists use a set of **perspectives** — lenses that ask different questions about the same behaviour. The best way to learn them is to point all seven at one problem. Take a top-performing salesperson who suddenly resigns.

Perspective	Focus	What it asks about the resignation
Psychodynamic	Unconscious motives, early experience	Is something they can't articulate driving this — a pattern of leaving when they feel controlled?
Behavioural	Reinforcement and punishment	Did the rewards stop? Was effort being punished with higher targets?
Humanistic	Growth, free will, self-worth	Did they feel stuck, with no room to grow?
Cognitive	Thinking, memory, decisions	How did they weigh the offer? What did they believe about their future here?
Sociocultural	Groups, culture, social norms	Did their peers leave? What does the team culture reward?
Biopsychological	Brain, hormones, physical health	Is chronic stress or sleep loss from travel wearing them down?
Evolutionary	Behaviour as adaptation	Does status loss inside the group trigger an exit, as it would in any primate hierarchy?

No lens is "right". A good diagnosis usually draws on two or three. The manager's mistake is to have only one — usually behavioural ("pay them more") or sociocultural ("bad team").

4 Who does what

A **psychologist** holds an academic degree in psychology and may do research, teaching, assessment or counselling. A **psychiatrist** is a medical doctor who specialised in mental disorders and can prescribe medication. In India, practising clinical psychologists must be registered with the Rehabilitation Council of India (RCI). The specialisations that touch business most are **I/O psychology** (selection, appraisal, training), **social psychology** (influence, groups, attitudes) and **consumer psychology**, which sits between psychology and marketing.

5 The scientific method in five steps

1. **Perceive a question.** "Do dealers who get more visits order more?"

2. **Form a hypothesis** — a specific, testable prediction. "Dealers visited weekly will place larger monthly orders than dealers visited monthly."
3. **Test it.** Collect data with a method suited to the question (sections 6–8).
4. **Draw conclusions.** Did the data support the hypothesis, and how strongly?
5. **Report the results** in enough detail that someone else can repeat the study. This step is where the field later stumbled (section 10).

Step 2 needs an **operational definition**: an exact statement of how a vague idea will be measured. "Dealer satisfaction" is vague. "Score on a five-question survey, 1–5 scale, collected by someone other than the dealer's own ASM" is operational. If you cannot write the operational definition, you do not yet have a hypothesis.

6 Describing behaviour: four descriptive methods

These methods tell you *what* is happening. None of them can tell you *why*.

- **Naturalistic observation** — watch behaviour where it naturally happens: shoppers in a paint store, a retailer talking to a customer. Realistic, but you control nothing, and people behave differently when they know they are watched (the **observer effect**). A mystery shopper is naturalistic observation done properly.
- **Laboratory observation** — observe in a controlled setting, such as an eye-tracking study of shelf layouts. More control, less realism.
- **Case study** — study one person, team or company in depth. Rich detail and good for generating ideas, but you cannot generalise from one case. Most business-school cases are exactly this, and the same limit applies.
- **Survey** — ask many people standard questions. Fast and scalable, but two problems are fatal if ignored. The **sample** must represent the population (surveying only dealers who answer the phone is not a sample of all dealers). And people do not always answer truthfully:

they give the socially acceptable answer, or the answer they think you want (**courtesy bias**). Ask a dealer whether he is satisfied with the company's service, in front of the company's ASM, and you have measured politeness.

7 Correlation: finding relationships

A **correlation** measures how strongly two variables move together. The **correlation coefficient, r**, runs from −1 to +1. The sign gives direction (positive: both rise together; negative: one rises as the other falls). The size gives strength (0 means no linear relationship; ±1 means a perfect straight line).

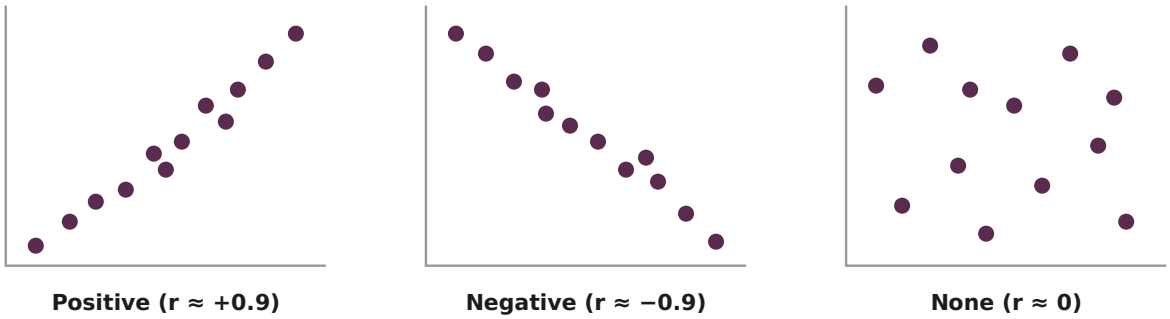


Figure 2. Three scatter plots. Each dot is one observation, such as one dealer.

WORKED EXAMPLE 1 · VISITS AND ORDERS

Five dealers in a territory. x = ASM visits per month, y = monthly order value (₹ lakh). Data invented for the exercise.

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Dealer      A     B     C     D     E      Mean
x (visits)  2     4     5     7     8      5.2
y (₹ lakh)  3     4     6     6     9      5.6

dx = x − 5.2   −3.2  −1.2  −0.2   1.8   2.8
dy = y − 5.6   −2.6  −1.6   0.4   0.4   3.4

Σ(dx·dy) = 8.32 + 1.92 − 0.08 + 0.72 + 9.52 = 20.4
Σdx²     = 10.24 + 1.44 + 0.04 + 3.24 + 7.84 = 22.8
Σdy²     =  6.76 + 2.56 + 0.16 + 0.16 + 11.56 = 21.2

r = Σ(dx·dy) / √(Σdx² × Σdy²) = 20.4 / √(22.8 × 21.2)
  = 20.4 / 21.99 ≈ +0.93
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A very strong positive correlation. The tempting conclusion is "more visits cause bigger orders, so double the visits". The data cannot support that.

Correlation is not causation

Three different worlds produce exactly the same r = +0.93:

1. Visits cause orders (what the ASM hopes).
2. Orders cause visits: ASMs, sensibly, spend time where the money is.
3. A **third variable** causes both. Bigger stores order more *and* get visited more, because they are important accounts.

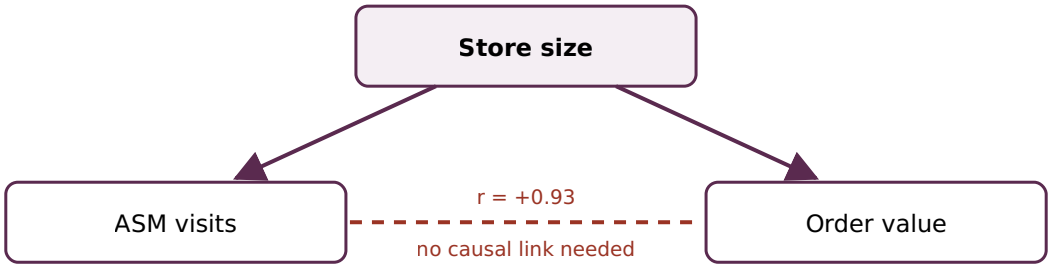


Figure 3. A third variable. Store size drives both visits and orders, so the two correlate even if visits change nothing.

Correlations are still valuable: they let you **predict** (high-visit dealers will probably keep ordering well) even when you cannot **explain**. Most of your dashboard is correlational. Read it that way.

8 The experiment: the only route to cause and effect

To show that one thing causes another, you must change it yourself and hold everything else constant. That is an **experiment**. Its parts:

- **Independent variable (IV)** — what the researcher deliberately changes (the incentive scheme).
- **Dependent variable (DV)** — what is measured to see the effect (order growth).
- **Experimental group** gets the IV; **control group** does not, or gets a **placebo** — something that looks like the treatment but isn't.
- **Random assignment** — each participant has an equal chance of landing in either group. This is the single most important feature: it spreads store size, owner temperament, location and every other hidden difference evenly across both groups, including the ones you never thought of.
- **Blinding** — in a **single-blind** study participants do not know their group; in a **double-blind** study the people collecting data don't either. This prevents the **placebo effect** (people improve because they expect to) and the **experimenter effect** (researchers unknowingly nudge results toward what they expect).

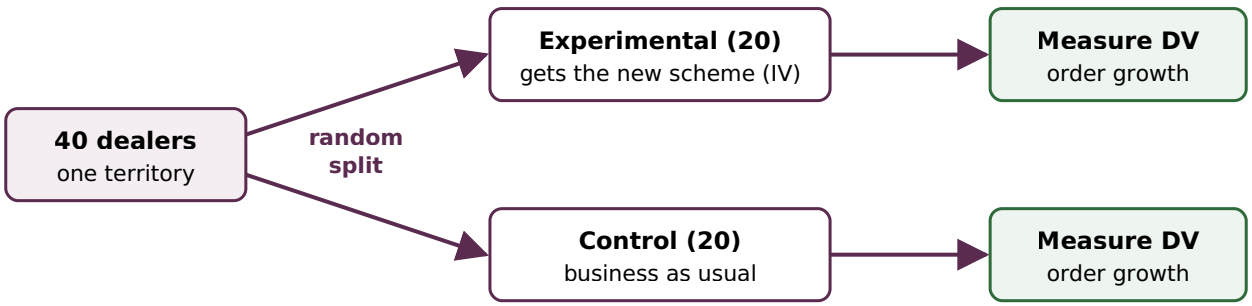


Figure 4. A field experiment. Business calls the same design an A/B test or a pilot with a control group.

WORKED EXAMPLE 2 · DOES THE SCHEME WORK?

Over one quarter the 20 scheme dealers grow orders by an average of 12%. The 20 control dealers grow by 7%. Same territory, same season, same products.

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Effect of scheme = 12% − 7% = 5 percentage points
Naive "before vs after" on scheme dealers alone = 12%
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Without the control group you would have credited the scheme with all 12 points. Seven of them would have happened anyway (season, market growth). The scheme's real effect is five — and only five is worth paying for.

The trap: suppose ASMs had *chosen* which 20 dealers got the scheme. They would pick their strongest, most responsive accounts. Now the groups differ before the scheme starts, and the 5-point gap could be entirely **selection bias**. This is why random assignment is not optional.

9 Ethics

Studies on people need **informed consent** (participants know what they are agreeing to and can quit), protection from unnecessary harm, confidentiality, and a **debriefing** afterwards that explains any deception. These rules exist because the field broke them. In the 1960s Stanley Milgram's obedience studies led participants to believe they were giving dangerous electric shocks to another person. Part 10 covers what those studies found. The point here: an employee "experiment" — a pilot scheme, a new appraisal format — affects real pay and careers. The same principles apply.

10 What's held up: the replication crisis

Step 5 of the scientific method — report so others can repeat — assumes someone actually repeats. For decades almost no one did. Journals rewarded new, surprising results; repeating someone else's study was unglamorous and hard to publish. Then the checks started.

In 2015 the **Open Science Collaboration**, a team of about 270 researchers, re-ran 100 studies published in three leading psychology journals in 2008. In the originals, 97% reported a significant result. In the repeats, only **36%** did, and the average effect was roughly half the original size. Your textbook, published in 2013, was written just before this landed.

Some famous findings that did not hold up:

- **Ego depletion** — the claim that willpower is a fuel tank that runs down with use. A 2016 replication across 23 labs and more than 2,000 participants found an effect close to zero.
- **Power posing** — standing in a "power pose" for two minutes supposedly raised confidence and changed hormone levels. Larger re-tests found no hormone effect, and one of the original authors publicly disowned the finding in 2016.
- **Mindless-eating research** — Brian Wansink's Cornell food lab produced widely-cited findings on plate sizes and portions. Cornell found him guilty of academic misconduct in 2018; he resigned, and many of his papers were retracted.
- **The Stanford Prison Experiment** (1971), long taught as proof that ordinary people turn cruel in the role of guard. Later archival work showed guards were coached toward harshness. It is now taught as a cautionary tale, not evidence.

Why did this happen? Small samples, which make flukes likely. Flexible analysis, where researchers try many tests and report the one that "worked" (**p-hacking**). And publication bias: studies that found nothing went in a drawer. The field's response has been real: **preregistration**

(publishing the hypothesis and analysis plan before collecting data), much larger samples, and journals accepting **registered reports** on design alone, before results are known.

WORKED EXAMPLE 3 · PRICING A CLAIM

A pre-2015 bestseller on "the science of persuasion" gives you ten striking lab findings. If they behave like the Open Science Collaboration sample, expect about **3 or 4** to replicate, and those at roughly half the advertised strength. Treat the book as a list of hypotheses to check, not a list of facts.

This is a rough prior, not a rule. Replication rates vary by subfield; cognitive findings held up better than social-psychology ones in the 2015 project.

SIX QUESTIONS BEFORE YOU BELIEVE A FINDING

1. Was it an experiment, or only a correlation?
2. How big was the sample? Dozens is weak; thousands is better.
3. Has an independent team replicated it?
4. Was it preregistered?
5. How large is the effect, not just whether it is "significant"?
6. Who paid for it, and who benefits if it's true?

From Part 1 onward, every major finding in this series gets a short **"Held up?"** note answering these questions.

Summary

- **Psychology** is the scientific study of behaviour and mental processes. Its four goals — describe, explain, predict, control — are also the four jobs of a manager diagnosing a problem.
- It began as a science in **1879** with Wundt's lab. Structuralism failed because introspection can't be checked. Functionalism, Gestalt, psychoanalysis and behaviourism each left ideas that survive; behaviourism's reinforcement is the basis of every incentive scheme.
- Seven modern **perspectives** (psychodynamic, behavioural, humanistic, cognitive, sociocultural, biopsychological, evolutionary) are lenses, not rivals. Use more than one.
- A **psychiatrist** is a medical doctor; a **psychologist** is not. I/O, social and consumer psychology matter most to business.
- The **scientific method**: question, hypothesis, test, conclude, report. A hypothesis needs an **operational definition**.
- **Descriptive methods** (observation, case study, survey) say what is happening, never why. Surveys fail on bad samples and courtesy bias.
- **Correlation** (r from −1 to +1) measures how two things move together and supports prediction. It never proves cause: reverse causation and third variables produce the same r.
- Only an **experiment** with **random assignment** and a **control group** shows cause and effect. Without a control, you credit your scheme with what would have happened anyway.
- Research on people requires consent, minimal harm and debriefing.
- Since 2015 the **replication crisis** has shown that many published findings — roughly two in three in one large check — do not repeat. Ask six questions before believing a claim: experiment or correlation, sample size, replicated, preregistered, effect size, who benefits.

Practice problems

P1. A brand manager stands in a paint retail store for a week, noting which shade cards shoppers pick up first, without telling them. Name the method. Give one strength and one weakness.

P2. HR finds that employees who attend more training sessions get better appraisal ratings (r = +0.40). The CHRO wants to make training mandatory to raise performance. Explain why the data doesn't support this, and name a plausible third variable.

P3. You want to know whether a WhatsApp payment reminder makes dealers pay on time. Design the experiment: state the IV, the DV, the control group, and how you would assign dealers.

P4. A territory's best retailer has stopped stocking your premium range. Give one question each from the behavioural, cognitive and sociocultural perspectives.

P5. Compute r by hand for four salespeople. x = years of experience: 1, 2, 3, 4. y = deals closed per month: 2, 4, 5, 7.

P6. A consultant pitches a training programme built on a 2011 study of 42 students showing that a two-minute pose raises confidence. Using the six questions, list three things you would ask before paying.

Glossary

Behaviour / mental processes
What can be observed / what goes on inside (thinking, feeling, remembering).

Perspective
A lens for explaining behaviour, e.g. behavioural or cognitive.

Hypothesis
A specific, testable prediction made before data is collected.

Operational definition
The exact procedure used to measure a concept.

Observer effect
People behaving differently because they know they are watched.

Courtesy bias
Giving the answer the asker seems to want, rather than the true one.

Correlation coefficient (r)
A number from −1 to +1 giving the direction and strength of a linear relationship.

Third variable
A hidden factor that causes both correlated variables.

Independent / dependent variable
What the experimenter changes / what is measured.

Random assignment
Placing participants in groups by chance, so groups start out equivalent.

Placebo effect
Improvement caused by expecting a treatment to work.

Double-blind
Neither participants nor data collectors know who is in which group.

Selection bias
Groups that differ before the treatment, usually because someone chose who went where.

Replication

Repeating a study, ideally by an independent team, to see if the result holds.

p-hacking

Running many analyses and reporting only the one that gave a significant result.

Preregistration

Publicly recording the hypothesis and analysis plan before collecting data.

Answer key

P1. Naturalistic observation. Strength: real behaviour in the real setting, and no observer effect if shoppers don't notice. Weakness: no control — you can't tell whether a card was picked for its colour or its position on the stand.

P2. It is a correlation, not an experiment. Possible third variable: motivation or conscientiousness — keen employees both sign up for training and work harder. Reverse causation is also possible: managers send their favoured performers to training. Test it by randomly assigning training to some employees.

P3. IV: receiving the WhatsApp reminder. DV: share of invoices paid by the due date (operational). Control: dealers who get no reminder (or a neutral message, as a placebo). Assignment: randomly, e.g. by coin-flip or random number per dealer, not by ASM choice.

P4. Behavioural: did the margins or schemes on premium stop rewarding him? Cognitive: does he believe premium won't sell to his customers? Sociocultural: are other retailers in the market dropping it, or is a competitor's representative closer to him?

P5. Means: $\bar{x} = 2.5$, $\bar{y} = 4.5$. dx = −1.5, −0.5, 0.5, 1.5; dy = −2.5, −0.5, 0.5, 2.5. $\Sigma(dx \cdot dy) = 3.75 + 0.25 + 0.25 + 3.75 = 8$. $\Sigma dx^2 = 5$, $\Sigma dy^2 = 13$. $r = 8 / \sqrt{65} = 8 / 8.06 \approx$ **+0.99**. Near-perfect — but four people is far too small a sample to trust.

P6. Any three: Has it been independently replicated? (Larger studies failed to.) n = 42 is small — any larger study? Was it preregistered? How large was the effect on actual job performance, not self-reported confidence? Who profits from the programme?

Sources: chapter structure follows Ciccarelli & White, *Psychology*, 4th ed. (Pearson, 2015), Ch. 1; text and figures are original. Replication figures: Open Science Collaboration, "Estimating the reproducibility of psychological science", *Science*, 2015; Hagger et al., "A multilab preregistered replication of the ego-depletion effect", *Perspectives on Psychological Science*, 2016. Worked-example data are invented for teaching.